Doc. 313 – No Beginning: The Thermal History on the Time Cycle

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Contents

Doc. 313 – No Beginning

What this is about

Doc. 312 ends with obligation D(ii): can the field configurations close periodically on the compact time cycle ($\tau_c = 2\pi/H_0 = 91.9$ Gyr) *while* carrying the observed thermal sequence – nucleosynthesis, recombination, structure formation? A circle has no beginning; the observations show a gradient. This document computes the question (`ffgft_313_zeitzyklus_geschichte.py`, `ffgft_313_fraktale_wegkorrektur.py`) and finds three results and one hard open core:

1. Ω_m **from** ξ **[K|P20**$\times E$**]:** the chain $H_0 \to T_0 \to \Omega_r \to$ antipode condition gives $\Omega_m^* = 0.3136$ against Planck 0.315 ± 0.007 ($-0.19\,\sigma$) – with no fitted parameter, since T_0 comes from the atomic scale (Ch. F).
2. **Antipode finding [K|P20]:** the observable history fills *one half-cycle*. With the fractal path correction (Ch. B, independently from A040) the particle horizon sits within $0.14\,\%$ of the antipode ($\theta = 0.9986\,\pi$; smooth: $1.012\,\pi$), the scattering wall at $0.979\,\pi$ – the "beginning" is the antipode of the time cycle, not an event.
3. **Smooth closure [K]:** the observed mass-drift profile reaches the antipode with slope $\sqrt{\Omega_r} \approx 0.01$ – the smoothness condition of an even periodic closure is met to $1\,\%$, and the return half is exactly the photonically unobservable half.
4. **Entropy obstruction [K]:** exact τ_c periodicity is not dynamically generic (Poincaré discrepancy $\sim 10^{10^{104}}$); it must be imposed as a boundary condition. Fine-grained admissible, coarse-grained open – the sharpened core of D(ii), which presents as a *fork*: Doc. 295 knows the non-closing case (log spiral) in which the entropy requirement lapses.

Not part of the A-series; registration in Doc. 190 separately. References: Doc. 306 ($\partial \mathrm{T}^4 = \emptyset$), 267 (degeneracy), 312 (closure scale, compact time direction, statics of the relation, torus rest frame).

Chapter A

No boundary, no $t = 0$: the starting position

FFGFT's position on singularity and big bang stands since Doc. 306: $\partial\mathrm{T}^4 = \emptyset$ – the torus has no boundary, "what was at $t = 0$?" is wrongly posed. Doc. 312 made this concrete and gave it three supports:

Geometric: the time direction is a cycle of length 91.9 Gyr. A closed circle has no starting point – "$t = 0$" is as meaningless on it as "the beginning of the equator".

Backwards: in the mass-drift reading (Doc. 312, statics of the relation; Wetterich), running back does not send distances to zero but masses: $\tilde{T} \cdot m = 1$ then means $\tilde{T} \to \infty$ – no density divergence, but a regime of ever slower clocks. The "singularity" is where the clocks stand still, not where the density diverges.

Forwards: no collapse into a singularity – the winding/momentum minimum (appendix of Doc. 312) has $V'' > 0$.

What Doc. 312 left open is the *history*: observations show a clear thermal sequence. How does a gradient live on a circle? That is D(ii), and here it is computed, not narrated.

Chapter B

Preliminary: the fractal path correction

All distances in this document are *light paths*. But FFGFT space is not smooth-affine; it carries the fractal dimension $D_f = 3 - \xi$ (A040). A traversed path accumulates an elongation over it, so the smoothly computed distance is too large. Before any numbers, therefore, the question of where this factor applies.

B.1 The factor

A040 supplies it, and *derived*, not stipulated:

$$K_{\text{frak}} = \left(\frac{D_f^{\text{eff}}}{3}\right)^{D_f^{\text{eff}}/2} = 1 - 100\,\xi = 0.98665, \qquad D_f^{\text{eff}} \approx 2.973. \tag{B.1}$$

D_f^{eff} is the unique value for which two independent derivations of m_e/m_μ agree – the factor 100 is thus a derived quantity, *not* a decade count and not something adjustable to a desired result. Its uncertainty is quantified: $n = 100 \pm 2$ (register 190, R67), i.e. $\pm 0.027\,\%$ in the effect. The path elongation is $1/K_{\text{frak}} - 1 = 1.35\,\%$.

B.2 The assignment rule

A040 restricts sharply [B]: K_{frak} concerns *absolute* values, never ratios *of the same level* – there it cancels. What “same level” means is decided by the distinction of Doc. 311:

Quantity	Level	K_{frak}
Light path η_0, $\chi(z)$	unrolled (traversed)	applies
L^*, R_H, τ_c	rolled up (topological)	—
path / structure ($\eta_0/\pi R_H$)	mixed	**applies**
path / path (χ_{rec}/η_0)	same level	cancels
angles, dimensionless ratios	same level	cancels

The reason is conceptual, not numerical: a light path exists only in the *unrolled* representation – in the rolled-up view there is no “path to the antipode”, only the cycle position θ. R_H, by contrast, follows from $H_0 = (\pi/2)c\xi^{10}/\bar{\lambda}_e$, a topological structural relation. So $\eta_0/(\pi R_H)$ compares something traversed with something topological and is *not* a same-level ratio.

Control test: $\chi_{\text{rec}}/\eta_0 = 0.98024$ is identical with and without the correction – two light paths, the factor cancels as required. The rule is thus not freely applicable; it has a testable invariant.

A second, independent test: the CMB peaks. The position of the first acoustic peak is $\ell_1 = \pi D_A/r_s$ – angular distance divided by sound horizon, both *traversed paths*. By the rule K_{frak} cancels, so the peak positions remain exactly unchanged. The cross-check shows how sharp this test is: had the factor applied to D_A alone, ℓ_1 would shift by $1.35\,\%$, i.e. by three multipoles – against a measured accuracy of ± 0.5 the rule would be falsified immediately. It passes this test without having been constructed for it.

Declaration: this assignment is a conceptual stipulation, and it improves the numbers. Both belong stated side by side. In its favour: the direction of the effect is physically compelled (a winding path covers less smooth distance in the same travel time – no choice), and the level distinction of Doc. 311 was formulated independently and earlier. Nothing known speaks against it, but it is not proven. All following numbers are therefore carried in both versions.

Chapter C

The antipode finding [K|P20]

C.1 Where does the history sit on the cycle?

Position on the cycle: $\theta = \chi/R_H$, with χ the light path distance (the $z \to \chi$ assignment takes the measured ΛCDM map; only dimensionless ratios enter, Doc. 267). The antipode lies at $\theta = \pi$, i.e. $\chi = L^*/2 = \pi R_H = 14.09$ Gpc. Computed:

	fraction of $L^*/2$		
	smooth	fractal	
Recombination ($z = 1090$)	0.992	0.979	−2.1 %
Particle horizon (z_{max}, Ch. D)	1.012	**0.9986**	**−0.14 %**

The **particle horizon** – the formal boundary of causal contact and hence the actual "beginning" – lands within 0.14 % of the antipode with the fractal correction, against 1.2 % in the smooth calculation. That is an improvement by a factor of 9 with nothing adjusted. The remainder is, however, five times the K_{frak} uncertainty (±0.027 %), so not an exact hit.

The **scattering wall** thereby moves from 0.992 to 0.979: the secondary finding "recombination at the antipode" becomes *weaker*. Both statements cannot be exact simultaneously – they differ by the invariant factor $\chi_{\text{rec}}/\eta_0 = 0.980$.

z	θ/π smooth	θ/π fractal	$m(z)/m_0$
0.5	0.140	0.138	0.667
1	0.243	0.240	0.500
3	0.465	0.459	0.250
10	0.689	0.680	0.091
100	0.917	0.905	0.0099
1090	0.992	0.979	0.00092

Finding: the entire observable history – up to $z \to \infty$ – fills one *half-cycle* to 1.2 %. The last-scattering surface sits at the antipode (99.2 %), the formal "beginning" 1.2 % behind it. The "big bang" is thereby a *place on the cycle* – the antipodal region, in the mass-drift reading the region of smallest masses and slowest clocks – and not an event before time.

C.2 Honest placement of the coincidence [S]

In ΛCDM language the same fact is the known quasi-coincidence $\eta_0 H_0/c \approx \pi$ (here 3.18 against 3.14, i.e. 1.2 %) – there an accident of the measured Ω values. In the compact reading it becomes a structural statement: *the scattering wall sits at the antipode.* For it to be more than a re-labelled coincidence, the location would have to follow forward. The fractal path correction is a step in exactly that direction: independently derived (A040), not fitted to this result, and it reduces the particle horizon's distance from the antipode from $1.2\,\%$ to $0.14\,\%$. The location is not thereby fully derived – the remainder is five times the R67 uncertainty, and why the matter content fills exactly the half-cycle remains open. Status [S], noted as a derivation target.

Chapter D

No infinity, no zero: the lower bound

Speaking of the "particle horizon at $z \to \infty$" is a loan from standard language and, in FFGFT, strictly speaking false: there are neither infinities nor zeros. The energy quantum of the time cycle (Doc. 312) is a hard *lower bound*:

$$E_{\text{min}} = \hbar H_0 = 2.28 \times 10^{-52}\ \text{J}, \qquad m_{\text{min}} = \frac{\hbar H_0}{c^2} = 2.54 \times 10^{-69}\ \text{kg}. \tag{D.1}$$

Its relation to the Kammerton is exactly the ladder step itself: $\hbar H_0/(m_e c^2) = (\pi/2)\,\xi^{10}$. From this follows a *finite* highest z: the mass drift ends when the electron mass reaches the quantum,

$$1 + z_{\text{max}} = \frac{m_e}{m_{\text{min}}} = \frac{1}{(\pi/2)\xi^{10}} = 3.59 \times 10^{38}. \tag{D.2}$$

D.1 What this changes numerically: nothing

The light-path contribution beyond $z = 10^9$ is $1/(\sqrt{\Omega_r}(1+z)) \approx 10^{-7}\,R_H$, beyond z_{max} still $10^{-37}\,R_H$. All numbers of the preceding chapters are unaffected.

D.2 What it changes conceptually: three things

(a) The singularity is excluded, not merely wrongly posed. So far the argument was: $\partial\mathrm{T}^4 = \emptyset$, the question about $t = 0$ is wrongly posed. Now a second, stronger one is added: masses *cannot* run to zero, because below $\hbar H_0/c^2$ no state exists. The antipodal region is not a limit one approaches but a reached endpoint of the scale.

(b) One scale sets both bounds. $\Lambda^* = 1/R_H^2$ bounds from above (largest length), $\hbar H_0$ from below (smallest energy) – the same ladder step ξ^{10}, no second parameter. The economy of the construction thus reaches beyond cosmology into quantisation.

(c) Poincaré presupposes finiteness. The recurrence theorem holds only for finite phase spaces. Without infinities and without a continuum down to zero, recurrence is therefore *guaranteed* – finiteness is the precondition of the cycle closure, not its obstacle. What remains open is only the time scale (Ch. E): that recurrence exists does not mean it happens within τ_c.

Chapter E

The smooth closure [K]

E.1 The profile and its endpoint slope

Periodicity requires the mass drift $m(\theta)$ to close continuously on the circle. The simplest closure is an *even* profile about the antipode: minimum at $\theta = \pi$, symmetric return on the other half. It is smooth if $dm/d\theta \to 0$ at the turning point.

In the radiation era $H \simeq H_0\sqrt{\Omega_r}(1+z)^2$, hence analytically

$$\eta_0 - \chi = \frac{c}{H_0\sqrt{\Omega_r}}\,\frac{1}{1+z} \qquad \Longrightarrow \qquad \left|\frac{dm}{d\theta}\right|_{\text{end}} = \sqrt{\Omega_r} \approx 0.0096.$$

Finding: the observed profile runs into the endpoint with slope ~ 0.01 – the smoothness condition of an even periodic closure is met to $1\,\%$. The deformation required relative to the measured history is a $1\,\%$ effect at the outermost edge.

E.2 Why this is not self-deception

The return half – where the mass drift would have to rise again – is *exactly* the half behind the scattering wall: photonically unobservable in principle. That is suspiciously convenient at first sight, but it is not a choice of this document; it is a consequence of the antipode finding: the wall *lies* at $0.992\,\pi$. The closure therefore contradicts no single photon observation – and it is nevertheless not untestable (Ch. E).

The climate-zone picture from the Doc. 312 discussion thereby becomes quantitative: the gradient is the map *along* the cycle (like climate zones along a meridian circle), not an evolution *out of* a beginning – and the "hottest" point of the map is the antipode.

Chapter F

The entropy obstruction: the core of D(ii) [K]

F.1 Periodicity is not generic

The entropy budget of the observable volume is $S \sim 10^{104}\,k_B$ (dominated by supermassive black holes; photons: $10^{89}\,k_B$). Poincaré recurrence of a generic state takes $\sim e^{S/k_B} \sim 10^{10^{104}}$ dynamical times – against $\tau_c \sim 10^{18.5}$ s. The discrepancy is $\sim 10^{104}$ *orders of magnitude in the exponent*: exact τ_c periodicity cannot be expected dynamically.

Consequence: periodicity is a **boundary condition** – a selection of periodic solutions from the solution space, not an outcome of evolution. It belongs to the same category as the choice of the T^4 topology itself ([STIPULATION] in the sense of Doc. 311) and must be booked as such.

F.2 Fine- and coarse-grained, separated

Fine-grained [K]: under unitary evolution the fine-grained entropy is constant; nothing microscopic forbids an exactly periodic solution. The boundary condition is admissible.

F.3 The fork: Doc. 295 knows the alternative

The account so far treats periodicity as the only option with an open entropy question. That is incomplete: Doc. 295 ("Time as a log spiral") already spells out the non-closure, and from the same factor. A winding that should advance by 2π per revolution advances only by πK_{frak}; the closure defect is

$$d = 100\,\xi = \frac{1}{75} \quad \text{per revolution.} \tag{F.1}$$

Three cases therefore stand side by side:

Case	Condition	Closure	Period
A	$d = 0$	after 1 revolution	91.9 Gyr
B	ξ frozen	after 75 revolutions	6892 Gyr
C	ξ running (recursion)	never	log spiral

Case B follows from the rational rotation number $1 - d = 74/75$; the $75 = 1/(100\xi)$ drops out exactly. Case C arises when ξ runs through the recursion ($\xi_n \approx 1/(100(n+75))$): the accumulated defect becomes logarithmic and the winding turns into the equiangular spiral $\rho = 75\, e^{\beta/2\pi}$ with self-similarity ratio e per precession round – not imposed but a consequence of the $1/n$ decay.

What this means for D(ii). Case A requires the coarse-grained closure after 91.9 Gyr – the open obligation. Case B relaxes it by a factor of 75 but changes nothing about the Poincaré discrepancy of 10^{104} dex in the exponent. **Case C removes the entropy requirement entirely**, because nothing has to return – but it costs the compact time direction in the sense of Doc. 312: τ_c, the quantum $\hbar H_0$ and the Gibbons–Hawking temperature would lose their basis.

The deciding question is thereby nameable and simpler than the entropy question itself: is the time winding bound to *one* scale step (then A or B), or does it run through the recursion (then C)? The recursion in Doc. 295 is a scale recursion, not a time recursion; whether the cosmological time direction traverses it is undecided. D(ii) is therefore more accurately carried as a *fork* than as a gap: two of the three routes are viable, but they lead to different theories.

Coarse-grained [open]: the second law is a statement about coarse-grained entropy relative to an observer. On the return half it must decrease – which is observer-relative possible (reversal of the correlation direction; no one *there* would see a violation, since the thermodynamic arrow turns with it), but unproven. **Exactly this is the sharpened core of D(ii):** not the field periodicity (which is consistent as a boundary condition, Ch. C), but the proof that a coarse-grained entropy closure is compatible with the observed structure formation. Until then: open, without embellishment.

Chapter G

Hawking radiation on the time cycle

Does Hawking radiation fit this picture? Yes – better than before, because the compact time direction (Doc. 312) has quietly supplied its foundation. At the same time it sharpens the entropy core of D(ii). Both computed (`ffgft_313_hawking_zyklus.py`).

G.1 One principle, three faces [K]

The three famous "temperatures out of nothing" – Unruh, Gibbons–Hawking, Hawking – are three separate miracles with three derivations in the textbooks. Behind them stands *one* KMS rule: where the time structure is periodic, an observer reads temperature, $T = \hbar/(k_B\tau)$:

System	Period τ	$T = \hbar/(k_B\tau)$
Cosmos (compact time)	$2\pi/H_0 = 2.9 \times 10^{18}$ s	2.63×10^{-30} K
Black hole ($M_\odot$)	$8\pi GM/c^3 = 1.24 \times 10^{-4}$ s	6.17×10^{-8} K
Unruh ($a = g$)	$2\pi c/a = 1.9 \times 10^8$ s	3.98×10^{-20} K

Doc. 312 *grounded* this rule for the cosmos: the Gibbons–Hawking temperature follows from the compactness of the time direction itself, without a horizon. Hawking radiation is therefore no foreign body and no extra postulate – it is the *local special case of the same rule*.

G.2 The membrane gets its thermometer [S]

The membrane picture (narrative Ch. 04: the hole *is* its fractal membrane, the radiation comes from the oscillating surface) so far had no answer to "why exactly this temperature?". Now it has one, in the language of the mass map: mass compresses the time structure ($\tilde{T} \cdot m = 1$ – the closer to the membrane, the slower the clocks), and at the membrane the local time structure becomes periodic with $\tau = 8\pi GM/c^3$. Short period, high temperature:

Object	T_H
PBH 10^{12} kg	1.2×10^{11} K
Earth	2.1×10^{-2} K
Sun	6.2×10^{-8} K
Sgr A*	1.5×10^{-14} K
SMBH $10^9 M_\odot$	6.2×10^{-17} K

The soap bubble of Ch. 04 has a thermometer, and the thermometer measures the clock map.

G.3 The family ladder [S]

Where does the ladder meet the cosmos? $T_H = T_{\text{GH}}$ requires $M = c^3/(4GH_0) = 4.66 \times 10^{52}$ kg – and its horizon lies at

$$r_s = \frac{2GM}{c^2} = \frac{R_H}{2} \quad \text{(exactly } 0.500\text{)},$$

at the same time exactly *half* the critical Hubble mass $c^3/(2GH_0)$. The temperature ladder runs from micro holes to the whole without a break: the cosmos is the largest family member.

G.4 What Hawking cannot do [K]

The honest friction with the cycle: evaporation times against $\tau_c = 91.9$ Gyr:

Object	t_{evap}	t_{evap}/τ_c
PBH 10^{12} kg	$10^{12.4}$ yr	$10^{1.5}$
Sun	$10^{67.3}$ yr	$10^{56.4}$
SMBH $10^9 M_\odot$	$10^{94.3}$ yr	$10^{83.4}$

Only primordial holes below the threshold mass

$$M^* = \left(\frac{\tau_c \, \hbar c^4}{5120\,\pi G^2}\right)^{1/3} = 3.3 \times 10^{11} \text{ kg} \quad (\sim \text{asteroid})$$

close via Hawking within one cycle. A solar-mass hole is 56 orders of magnitude too slow, an SMBH 83. The FFGFT correction from Ch. 04, $P = P_{\text{std}}(1 - \xi \ln(M/M_P))$, amounts to 0.6–$1.4\,\%$ and changes nothing in this finding.

Consequence for D(ii): stellar and supermassive holes – the carriers of the 10^{104} entropy budget of Ch. D – cannot disappear via Hawking on the cycle. The return half must dismantle them by another route. Hawking radiation thus fits the picture thermodynamically perfectly and at the same time *sharpens* the coarse-grained core of D(ii): it shows how hopelessly slow the only known return route would be.

G.5 Information: two languages, one statement [S]

Ch. 04 always said: the radiation correlates with the fractal membrane structure – nothing is lost ($S_{\text{BH}}(M_\odot) = 1.05 \times 10^{77}\, k_B$ of correlations). Ch. D of this document says: fine-grained entropy is constant under unitary evolution – periodicity is microscopically admissible. That is the same statement in two languages: information preservation locally (membrane) and globally (cycle) are consistent. What remains open is solely the *coarse-grained* closure – which was the core all along.

Chapter H

Test surfaces [S]

Photons end at the wall ($0.992\,\pi$). Two messengers originate at or behind the antipode:

Messenger	Decoupling	Position
CνB (neutrino background)	$z \sim 6 \times 10^9$	$\theta \approx 1.01\,\pi$
primordial GW	nominally $z \sim 10^{25}$	$\theta \approx 1.01\,\pi$

What the path correction costs here: fractally, the scattering wall lies at $0.979\,\pi$, clearly *below* the half-cycle – it does not intersect itself. On the one hand this makes the compact reading more compatible with the null results of the topology searches (the bound $(L^*/2)/D_{\mathrm{LSS}}$ rises from 1.008 to 1.022); on the other it removes a test signature: without self-intersection there are no matched circles and hence no $4.2'$ offset (Doc. 312). The price of the better numbers is a lost test opportunity – to be booked honestly, not concealed.

A *non-monotonic* signature there – the reversal of the mass drift beyond the antipode – thus remains the *only* surviving test surface of the closure.

H.1 The low-ℓ cutoff: a second test surface [K|P20]

There is a second one, and it lies closer to existing data. On a torus of edge length L^* no modes with wavelength $> L^*$ exist; the smallest admissible wavenumber is $k_{\mathrm{min}} = 2\pi/L^*$. Translated into angular scale:

$$\ell_{\mathrm{min}} = k_{\mathrm{min}}\, D_{\mathrm{LSS}} = \frac{2\pi D_{\mathrm{LSS}}}{L^*} = 3.08. \tag{H.1}$$

Below $\ell \approx 3$ modes are missing: the quadrupole ($\ell = 2$) is strongly suppressed, the octupole ($\ell = 3$) sits at the edge, and from $\ell \geq 4$ the spectrum is unaffected.

No fitted parameter. $L^* = 4\bar{\lambda}_e/\xi^{10}$ follows from the chain, D_{LSS} from the same one; varying Ω_m between 0.305 and 0.325 moves ℓ_{min} only from 3.12 to 3.04. Without the fractal path correction it would be 3.12 – the statement is robust against both freedoms.

What is observed. Since COBE (1992), confirmed by WMAP and Planck: the quadrupole lies at about one sixth of the ΛCDM expectation, the octupole slightly low, and from $\ell = 4$ the spectrum is normal. In ΛCDM this has been unexplained for three decades and is attributed to cosmic variance. The compact reading predicts exactly this structure – *one* number, in the right place, with an intact spectrum above it.

What is still missing. The precise degree of suppression requires the mode summation on the torus instead of the integration over $\mathbb{R}^3$ – a finite sum, not a Boltzmann code, but not yet carried out. Only that decides whether the factor $1/6$ is matched quantitatively or only in direction. Until then the agreement counts as an *order statement* [K|P20], not a precision test. *Check script:* `ffgft_313_lowl_cutoff.py`. Practically beyond present measurement, but nameable; together with the $4.2'$ dipole offset (Doc. 312, torus rest frame) these are two concrete signatures of the compact reading.

Chapter I

The antipode condition as a prediction: Ω_m^* and the κ candidate

I.1 The inversion

Ch. B read the location of the scattering wall as a finding and honestly booked the coincidence $\eta_0 H_0/c \approx \pi$ as [S]. But the condition can be *inverted*: if the compact reading requires the particle horizon to close exactly at the antipode,

$$\eta_0 = \pi R_H \qquad \Longleftrightarrow \qquad \int_0^\infty \frac{dz}{E(z)} = \pi, \tag{I.1}$$

then Ω_m is no longer a free parameter. The condition solves uniquely (flat, $\Omega_r = 9.3 \times 10^{-5}$) to:

Without the fractal correction this gives $\Omega_m^* = 0.325$ ($+1.4\,\sigma$ vs Planck). With it, the condition reads $\int dz/E = \pi/K_{\text{frak}} = 3.1841$ and yields $\Omega_m^* = 0.314$ ($-0.2\,\sigma$). The R67 uncertainty ($n = 100 \pm 2$) shifts this only between 0.3137 and 0.3141. The next section traces the chain back to ξ entirely.

The cosmic sector thereby obtains, for the first time, a *prediction for a ΛCDM parameter* rather than a reinterpretation – falsifiable: sharper Ω_m measurements (DESI, Euclid) decide whether 0.325 holds.

What the comparison quantity is: Planck $\Omega_m = 0.315 \pm 0.007$ is not a raw datum but itself the output of a six-parameter ΛCDM fit to photon count rates – including ΛCDM transfer functions and fiducial cosmology in the analysis chains. Read precisely, the $1.4\,\sigma$ comparison is therefore: *condition against pipeline, not condition against nature.* This is the third inheritance level beside the SI anchor (Doc. 309) and the $E(z)$ form (above), and like those it is declared rather than hidden. It cuts both ways: decades of ΛCDM refinement produce robustness *and* pipeline lock-in – and where the compact reading inherits the form, it simultaneously inherits its fit successes for the shared parts. This is why the independent test points of this document deliberately sit in the minimally model-laden observable class: directions and shapes (the $4.2'$ dipole offset, non-monotonicity at the antipode) rather than fit parameters.

I.2 The chain: prediction from ξ alone

Decisively, T_0 here does *not* come from cosmology: Doc. 061 derives it from an atomic energy scale ($k_B T_0 = (16/9)\xi$, Sec. F.3). The chain is therefore circle-free and hangs on a single number:

	Step	Value
(1)	$H_0 = (\pi/2)\,c\,\xi^{10}/\bar{\lambda}_e$	$h = 0.6682$
(2)	$k_B T_0 = (16/9)\,\xi$ (Doc. 061)	$T_0 = 2.751$ K
(3)	$\Omega_r \propto T_0^4/h^2$	9.643×10^{-5}
(4)	$K_{\text{frak}} = 1 - 100\,\xi$ (A040)	0.98667
(5)	$\int dz/E = \pi/K_{\text{frak}}$	$\boxed{\Omega_m^* = 0.3136}$

against Planck 0.315 ± 0.007: $-0.19\,\sigma$. Using the *measured* instead of the derived T_0 gives 0.3139 ($-0.16\,\sigma$) – the prediction is practically insensitive to the $0.92\,\%$ deviation of the T_0 formula (it propagates as 0.0003). No step of the chain contains a parameter fitted to Ω_m.

I.3 T_0 overdetermined: two sectors, one value

Two mutually independent routes lead to T_0, and both use fixed natural scales – no free parameter:

- **Route 1 (particle sector):** $k_B T_0 = (16/9)\,\xi$ from Doc. 061 – tied to the atomic energy scale, with no cosmological input.
- **Route 2 (cosmic sector):** the antipode condition, tied to H_0 via the Kammerton $\bar{\lambda}_e$ – with no particle-physics input.

T_0 is thereby *overdetermined*, and the agreement is testable. The direction of leverage matters:

Ω_m	T_0 from route 2
0.3100	3.062 K
0.3136	2.754 K
0.3150	2.625 K
0.3200	2.097 K

The leverage is $dT_0/T_0 \big/ d\Omega_m/\Omega_m \approx -12$: one per cent in Ω_m propagates as twelve per cent in T_0. Route 2 is therefore a sharp Ω_m determiner and a blunt T_0 determiner – the comparison must be made on the Ω_m side, not the T_0 side.

Read that way, the overdetermination holds: the model-independently measured $T_0 = 2.7255$ K (FIRAS, $0.02\,\%$) gives $\Omega_m = 0.3139$ via the antipode condition; the particle sector via route 1 gives $\Omega_m = 0.3136$. *Two independent sectors, the same number to* $0.1\,\%$ – the same pattern as for α in A130, where two routes to E_0 coincide.

Consequence for the status: the chain of the previous section is thus less a prediction than a *consistency check between the particle and the cosmic sector* – and that is the stronger statement, because it can break. The test channel runs through Ω_m: a determination to $0.3\,\%$ (DESI, Euclid) sharpens it by a factor of 2.

Declared inheritance: the calculation uses the ΛCDM *form* of $E(z)$ – including the $(1{+}z)^3$ term with full Ω_m. The prediction is therefore stated precisely: *the map must be shaped so that* $\int dz/E = \pi$*; in* Λ*CDM bookkeeping this reads* $\Omega_m = 0.325$. What carries the matter share of that bookkeeping is a separate question (Sec. F.3). Status hence [K|P20 $\times\, E(z)$ form] – the same inheritance structure as in Doc. 309, one level deeper.

I.4 Consolidation: obligation B and the coincidence become one condition

From Ω_m^* follows $\Omega_\Lambda^* = 0.675$ and hence a structural candidate for the so far open order-one factor of the closure scale (Doc. 312, obligation B):

$$\kappa^* = 3\,\Omega_\Lambda^* = 2.058 \qquad \text{against measured } \kappa = 2.12 \;\; (-2.9\,\%) \tag{I.2}$$

(smooth: 2.026, i.e. $-4.4\,\%$).

The same consolidation logic as with the 10^{123} problem (→ P20): two open items – obligation B and the [S] coincidence of Ch. B – are reduced to *one* condition. The cross-check shows this is not circular: $\kappa = 2.12$ would require $\Omega_m = 0.294$, and then $\eta_0 H_0/c = 3.268 \neq \pi$. Antipode exactness and today's Planck κ stand in a genuine $4\,\%$ tension – test contact, not a fit.

I.5 Placements and one new open item

Ω_Λ loses its substance status: the "dark-energy density" is the FLRW translation of the closure scale; if the antipode condition holds, it is predicted, not fitted. **Hubble tension:** the compact reading takes an unambiguous side with the CMB-anchored value ($H_0 = 66.8$ from the ξ^{10} chain against Planck 67.4); the local SH0ES excess would then have to be a map effect of the calibration [S].

Dark matter: two regimes, one open. Dark *matter* has deliberately not been mentioned so far – and the gap belongs named. *Galactically* (rotation curves, RAR) FFGFT needs no particle DM: there the transition scale a_0 carries the load (Doc. 308); this document says nothing about that, correctly so. *At the background level*, however, the $z \to \chi$ map silently inherits the CDM share: in ΛCDM bookkeeping, $\Omega_m^* = 0.325$ consists of baryons (≈ 0.049) plus cold dark matter (≈ 0.276). What carries this share in the compact reading – actual matter, or a property of the mass-drift profile $m(\theta)$ that books as matter in FLRW translation – is unresolved and is booked as a **second new open item**. It is independent of the galactic question and must not be conflated with it.

T_0: bound, and the unit question is settled. The corpus contains a forward formula (Doc. 061): $T_{\mathrm{CMB}}/E_\xi = (16/9)\xi^2$ with $E_\xi = 1/\xi$, i.e. $k_B T_0 = (16/9)\,\xi$ in electronvolts $= 2.370 \times 10^{-4}$ eV against the measured 2.349×10^{-4} eV, a hit to $+0.92\,\%$; the factor $16/9 = (4/3)^2$ is the same structure as $\xi = (4/3) \times 10^{-4}$. In addition Doc. 279 binds T_0 relationally to the Casimir length $L_\xi = 100.24\ \mu$m.

The unit is not an oversight but the reference scale of the whole sector. A266 [K] states it explicitly for $\alpha = \xi E_0^2$: the equation is correct only as $\alpha = \xi (E_0/1\,\text{MeV})^2$; the 1 MeV in the denominator is the *implicit reference scale*, which "does not appear explicitly in the shorthand but is indispensable", and its choice is "not free – it is the empirical SI anchor of the FFGFT sector". The eV notation in 061 is the same convention, merely written one order of magnitude smaller. The ratio $1\,\text{eV}/E_0 = 1.35 \times 10^{-7}$ is therefore the quotient of two *reference scales*, not a physical quantity; demanding an integer power of ξ from it would make as much sense as demanding one from 1 inch/1 metre.

What follows for the status. T_0 is *overdetermined* (Sec. I.3: two independent sectors, $0.1\,\%$) and bound to the atomic scale via 061. There is no separate open item "T_0". What remains is the question of the *location of the anchor itself* – why the reference scale lies where it lies – and that is identical with P20 and the inheritance problem of Doc. 309, hence already booked.

Register candidates (two). *(i) Notation clash:* in A085, T_0 in the four-garments identity $m_0 c^2 = E_0 = k_B T_0 = \hbar/\tilde{T}$ denotes the *calibration temperature* (8.53×10^{10} K), whereas in

Docs. 061/313 it denotes the *CMB monopole* (2.7255 K) – two quantities a factor 3×10^{10} apart under the same symbol. *(ii) Precision claim:* Doc. 061 states "$0.0002\,\%$" in its summary while its own detailed calculation reports "agreement to $1\,\%$"; the correct value is $0.92\,\%$.

Chapter J

Status overview

Statement	Content	Status
Statement	Content	Status
Antipode finding	horizon at antipode: fractal $0.14\,\%$ (smooth $1.2\,\%$); wall $0.979\,\pi$	[K|P20]
Path correction	$K_{\mathsf{frak}} = 1 - 100\xi$ applies to path/structure, cancels for path/path (test passed)	[K]/[S]
Coincidence placement	$\eta_0 H_0/c \approx \pi$: structural statement only after forward derivation	[S]
Smooth closure	endpoint slope $\sqrt{\Omega_r}/K \approx 0.0098$; closure met to $1\,\%$	[K]
Return half	= photonically unobservable half; no conflict with photon data	[K]
Lower bound	$\hbar H_0$ as smallest energy: no $z \to \infty$; singularity excluded by quantisation	[K]
Periodicity	boundary condition (Poincaré discrepancy 10^{104} dex in the exponent), not dynamics	[K]
Fork	three cases (Doc. 295): $d = 0$ / 75 revolutions / spiral; C removes the entropy requirement but costs τ_c	[K]
Entropy	fine-grained admissible; coarse-grained closure open – sharpened core of D(ii)	open
Hawking	local case of the same KMS rule (GH from Doc. 312); ladder meets cosmos at $r_s = R_H/2$	[K]/[S]
Evaporation	$M^* = 3.3 \times 10^{11}$ kg; $M_\odot$: 56 dex too slow – sharpens D(ii)	[K]
Test surfaces	CνB/GW from the antipode; non-monotonicity as signature	[S]
Low-ℓ cutoff	$\ell_{\mathsf{min}} = 2\pi D_{\mathsf{LSS}}/L^* = 3.08$; matches the quadrupole anomaly open since COBE	[K|P20]
Ω_m^*	chain from ξ alone: 0.3136 ($-0.19\,\sigma$); $E(z)$ form inherited	[K|P20×E]
Overdetermination	particle sector 0.3136 vs. cosmic sector 0.3139: consistent to $0.1\,\%$	[K]
κ^*	fractal 2.058 ($-2.9\,\%$), smooth 2.026: candidate for obligation B	[S]
Matched circles	wall without self-intersection: signature from Doc. 312 lapses	loss

Statement	Content	Status
DM (background)	$(1+z)^3$ share of the map: carrier unresolved (galactic: a_0, Doc. 308, separate)	open
T_0	*overdetermined* (two sectors, $0.1\,\%$); eV notation $=$ reference scale per A266, no separate open item	[K]

Balance: after this document, “no big bang” means precisely: the “beginning” is the antipode of the time cycle – a place, not an event; the observed history is one half of a map closable smoothly to $1\,\%$ (with the fractal path correction the particle horizon meets the antipode to $0.14\,\%$); and the remaining burden lies not with the fields but with coarse-grained entropy. D(ii) is thereby not settled but reduced to its hard core.

Registration: entry into Doc. 190 will be done separately.